# Data Interpretation Exercise

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**Question 1: Mapping Data onto Aesthetics**

**1. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.**

```
library(ggplot2)

# Dataset
sales <- data.frame(
  Product = c("P1","P2","P3","P4","P5","P6"),
  Sales = c(120,90,150,80,135,100),
  Profit = c(18,12,25,8,22,16),
  Category = c("Electronics","Clothing","Electronics",
          "Furniture","Furniture","Clothing")
)

# Scatter Plot
ggplot(sales, aes(x = Sales,
           y = Profit,
           color = Category,
           size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(title="Sales vs Profit",
     x="Sales (₹ Lakhs)",
     y="Profit (%)") +
  theme_minimal()
```

**2. Map Category using an appropriate aesthetic and Sales using another suitable aesthetic.**

**3. Interpret the relationship between Sales and Profit.**

From the scatter plot, the points show an upward trend. Products with higher sales generally have higher profit percentages. For example, P3 has the highest sales (150) and also the highest profit (25%), while P4 has the lowest sales (80) and the lowest profit (8%). This indicates a positive relationship between Sales and Profit.

**4. Identify the type of scale used for each variable.**

| Variable | Scale Type |
|---|---|
| Sales | Continuous |
| Profit | Continuous |
| Category | Categorical (Discrete) |

aes(

x = Sales,

y = Profit,

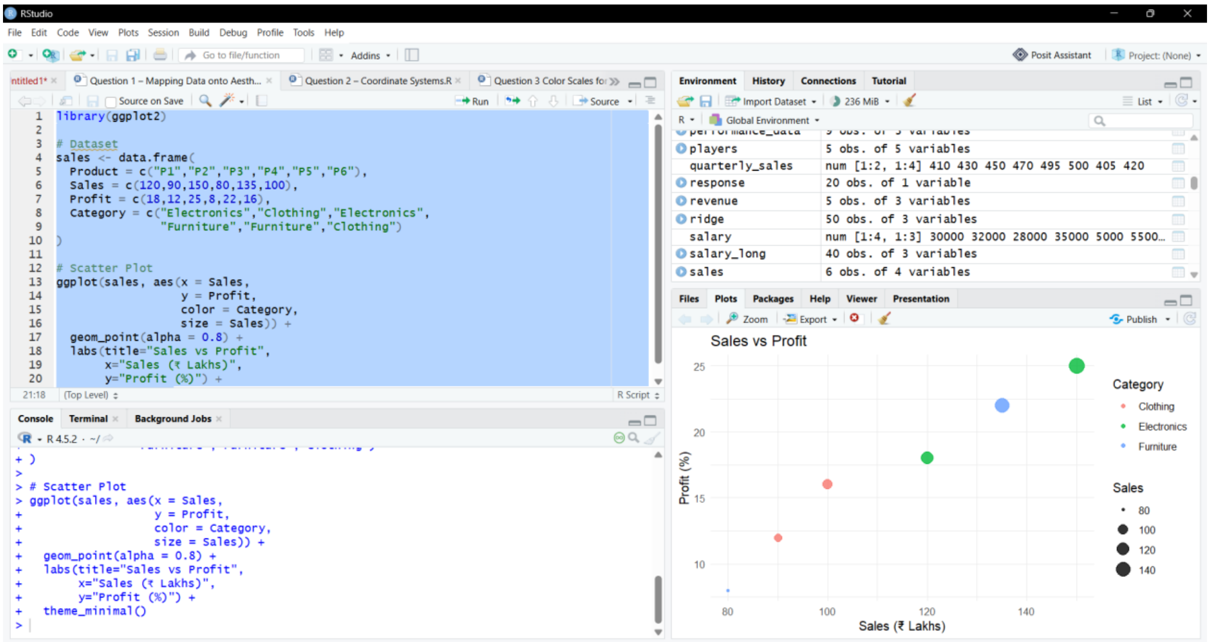
color = Category,

size = Sales

)

**5. Suggest one alternative visualization that better represents the same data.**

An alternative visualization is a **Bubble Chart**, which can effectively display Sales, Profit, and Category simultaneously.




## Question 2: Coordinate Systems (Cartesian and Polar)

1. **Write an R program to visualize the dataset using a Cartesian coordinate system.**

```
library(ggplot2)

direction <- data.frame(
 Direction=c("N","NE","E","SE","S","SW","W","NW"),
 Frequency=c(18,12,20,15,22,11,16,14)
)

ggplot(direction,
    aes(Direction, Frequency,
      fill=Direction)) +
 geom_bar(stat="identity") +
```

```
 labs(title="Direction Frequency (Cartesian)") +
 theme_minimal()



library(ggplot2)


ggplot(direction,
    aes(x="", y=Frequency,
      fill=Direction)) +
 geom_bar(stat="identity",
      width=1) +
 coord_polar(theta="y") +
 labs(title="Direction Frequency (Polar)") +
 theme_void()
```

2. **Write another R program to visualize the same data using a Polar coordinate system.**

The same dataset is visualized using a **Polar coordinate system** with coord_polar(), producing a circular chart.

**3. Compare the effectiveness of both visualizations.**

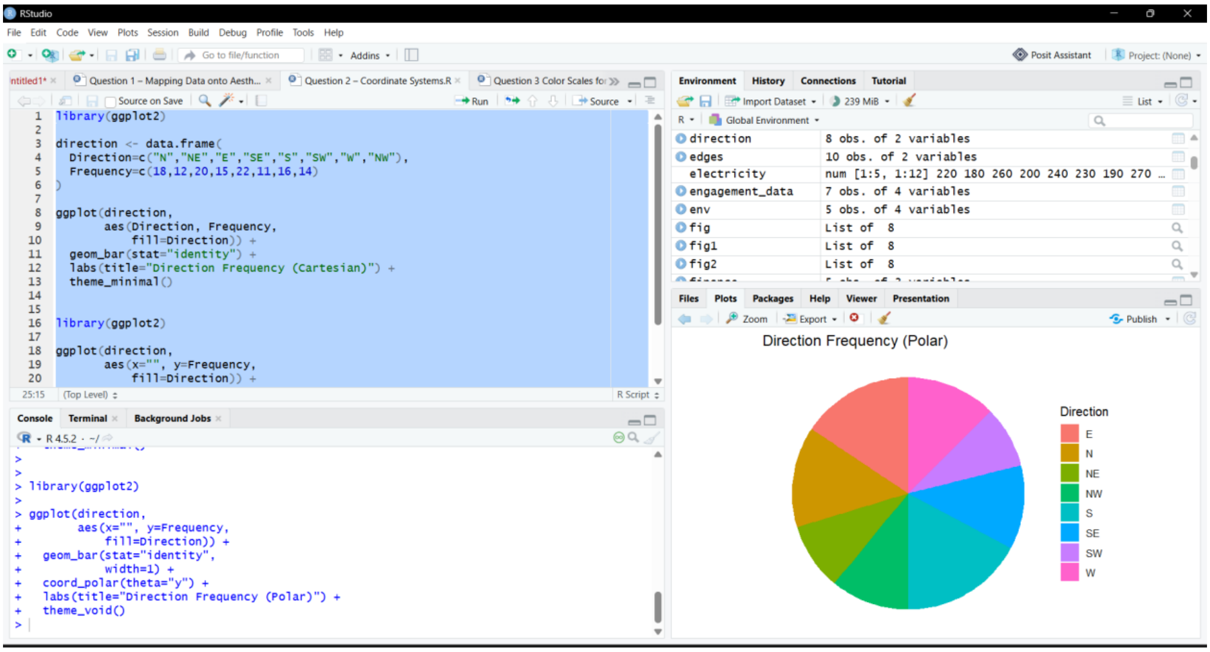
The Cartesian chart is better for comparing exact frequencies, whereas the Polar chart is more suitable for showing circular or directional patterns.

4. **Explain which coordinate system is more appropriate for directional data.**

The **Polar coordinate system** is more appropriate for directional data because compass directions naturally follow a circular arrangement.

5. **Justify your answer based on data interpretation.**

The Polar chart provides a better representation of directional information, making it easier to interpret the relationship between different compass directions.




## Question 3: Color Scales for Data Representation

**1. Write an R program to create a bar chart using a sequential color scale based on Marks.**

```
library(ggplot2)

student <- data.frame(
 Student=c("A","B","C","D","E","F"),
 Marks=c(45,58,72,81,67,90)
)

ggplot(student,
    aes(Student,
      Marks,
      fill=Marks)) +
 geom_bar(stat="identity") +
 scale_fill_viridis_c() +
 labs(title="Student Marks") +
 theme_minimal()
```

**2. Identify the students with the highest and lowest marks from the visualization.**

Highest Marks: Student F (90)

Lowest Marks: Student A (45)

3. **Explain why a sequential color scale is appropriate.**

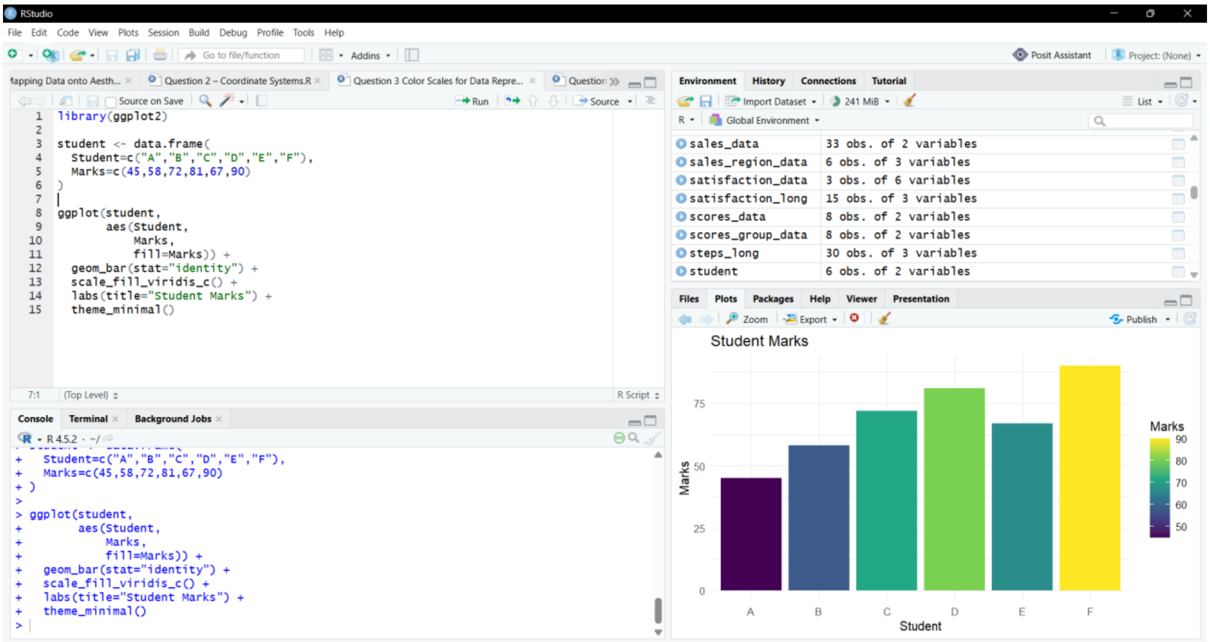
   A sequential color scale is appropriate because **Marks** is a continuous numerical variable, and the color intensity represents increasing or decreasing values effectively.

4. **Suggest a color palette suitable for color-blind users.**

A suitable color palette for color-blind users is the Viridis color palette.

5. **Explain the difference between sequential and categorical color scales.**

| **Sequential Color Scale** | **Categorical Color Scale** |
|---|---|
| Used for continuous numerical data | Used for categorical data |
| Represents low to high values | Distinguishes different groups using separate colors |




**Question 4: Highlighting with Color**

**1. Write an R program to create a bar chart highlighting the region with the highest revenue.**

```
library(ggplot2)

revenue <- data.frame(
  Region=c("North","South","East","West","Central"),
  Revenue=c(820,760,790,1010,770)
)

revenue$Highlight <- ifelse(revenue$Revenue==max(revenue$Revenue),
                    "Highest","Others")

ggplot(revenue,
      aes(Region,
        Revenue,
        fill=Highlight)) +
  geom_bar(stat="identity") +
```

```
scale_fill_manual(values=c("Highest"="red",
                 "Others"="steelblue")) +
labs(title="Regional Revenue") +
theme_minimal()
```

2. **Explain how highlighting improves data interpretation.**

   Highlighting improves data interpretation by drawing immediate attention to the most important data point.

3. **Identify the highest-performing region.**

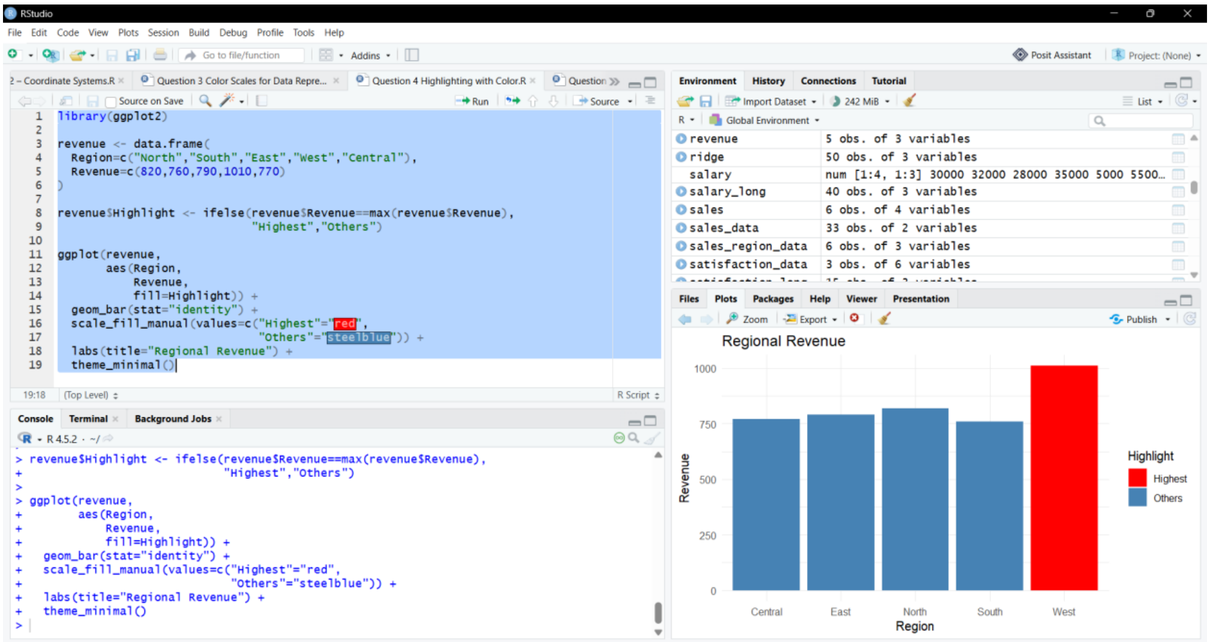
   The **West** region has the highest revenue of **₹1010 Crores**.

4. **Discuss how excessive use of colors can affect interpretation.**

Using too many colors can confuse the viewer, reduce readability, and make important information difficult to identify.

5. **Suggest one alternative highlighting technique other than color.**

   An alternative highlighting technique is to use **data labels (annotations)** to emphasize the highest value instead of color.




## Question 5: Integrated Visualization

1. **Write an R program to create a bubble chart using appropriate aesthetic mappings.**

```
library(ggplot2)

city <- data.frame(
 City=c("A","B","C","D","E"),
 Population=c(8.2,5.6,9.4,6.8,4.9),
 AQI=c(110,180,90,240,150),
 Rainfall=c(850,430,1200,520,680),
 Region=c("North","South","North","East","West")
)

ggplot(city,
    aes(x=Population,
      y=AQI,
      size=Rainfall,
      color=Region)) +
 geom_point(alpha=0.7) +
 labs(title="Population vs AQI",
```

```
    x="Population (Million)",
    y="AQI") +
 theme_minimal()
```

2. **Select suitable aesthetics for Population, AQI, Rainfall, and Region.**

| Variable | Aesthetic Mapping |
|---|---|
| Population | X-axis |
| AQI | Y-axis |
| Rainfall | Bubble Size |
| Region | Color |

3. **Interpret the relationship between Population and AQI.**

The visualization suggests that cities with larger populations may have higher AQI values in some cases, but the relationship is not perfectly linear.

4. **Identify the continuous and categorical variables.**

**Continuous Variables:**

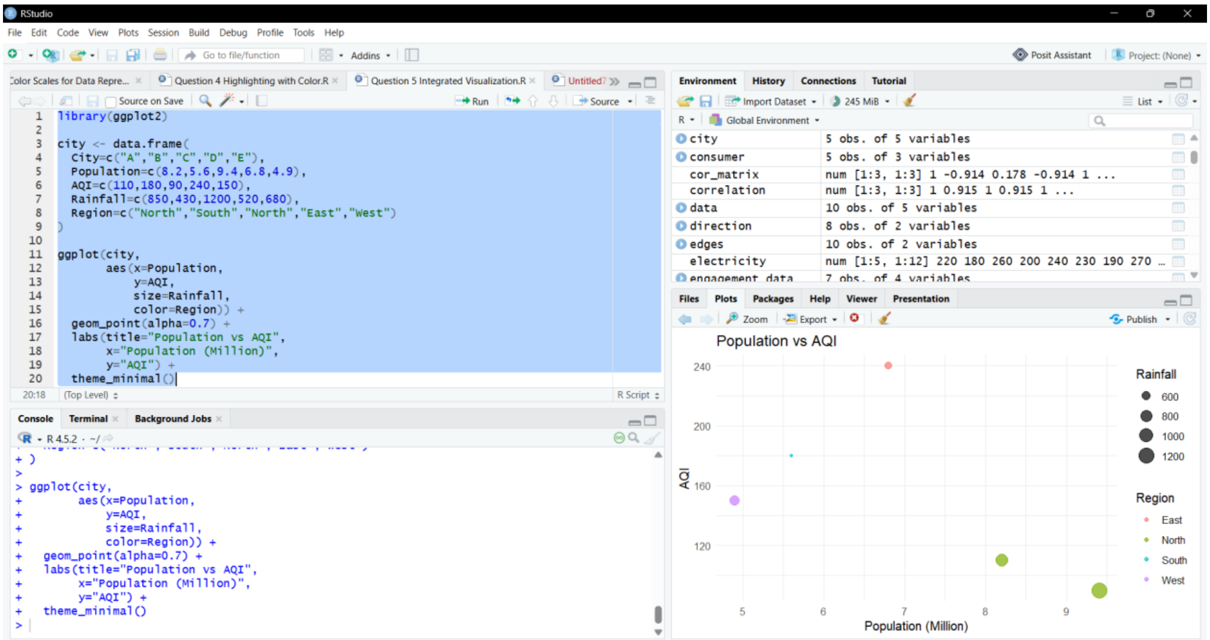
- Population
- AQI
- Rainfall

**Categorical Variables:**

- City
- Region

5. **Recommend one improvement to enhance the visualization's readability.**

A useful improvement is to add **data labels (geom_text())** or increase transparency to make the bubbles easier to identify and improve readability.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
Color Scales for Data Repre...
Question 4 Highlighting with Color.R
Question 5 Integrated Visualization.R
Untitled7
Source on Save
Run
Source
library(ggplot2)
city <- data.frame(
City=c("A","B","C","D","E"),
Population=c(8.2,5.6,9.4,6.8,4.9),
AQI=c(110,180,90,240,150),
Rainfall=c(850,430,1200,520,680),
Region=c("North","South","North","East","West")
)
ggplot(city,
aes(x=Population,
y=AQI,
size=Rainfall,
color=Region)) +
geom_point(alpha=0.7) +
labs(title="Population vs AQI",
x="Population (Million)",
y="AQI") +
theme_minimal()
20:18
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.5.2 · ~/
+ )
>
> ggplot(city,
+ aes(x=Population,
+ y=AQI,
+ size=Rainfall,
+ color=Region)) +
+ geom_point(alpha=0.7) +
+ labs(title="Population vs AQI",
+ x="Population (Million)",
+ y="AQI") +
+ theme_minimal()
>
Environment
History
Connections
Tutorial
Import Dataset
245 MiB
List
Global Environment
city 5 obs. of 5 variables
consumer 5 obs. of 3 variables
cor_matrix num [1:3, 1:3] 1 -0.914 0.178 -0.914 1 ...
correlation num [1:3, 1:3] 1 0.915 1 0.915 1 ...
data 10 obs. of 5 variables
direction 8 obs. of 2 variables
edges 10 obs. of 2 variables
electricity num [1:5, 1:12] 220 180 260 200 240 230 190 270 ...
engagement_data 7 obs. of 4 variables
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Population vs AQI
240
200
160
120
AQI
5
6
7
8
9
Population (Million)
Rainfall
600
800
1000
1200
Region
East
North
South
West